



Today, **April 7, 2026**, we've made significant strides in upgrading the **Monetchat/Sell-Sell Arabic Marketplace** experience. The core focus was on transitioning from AI-generated, "robotic" Arabic into a high-quality, deterministic, and premium conversational engine tailored for **Gulf (GCC) users**.



Summary of Key Updates

1. Strategic Shift to Model-Driven Arabic Engine

Designed and implemented a new architecture that separates **AI Reasoning** from **Arabic Voice**.

The Reasoning Engine (AI): Now focuses strictly on detecting user intent and extracting key entities (brands, models, prices, locations).

The Arabic Response Engine (Logic): Takes the AI's data and builds perfect, deterministic Arabic sentences that never drift in quality.

2. Deterministic Arabic Engine (



src/lib/ai/arabic-engine.ts)

Created a new core module that centralizes all user-facing Arabic.

Dynamic Phrasing: Responses now automatically include result counts and keywords (e.g., "لقيت لك 5 من خيارات آيفون").

Confidence Layer: Added a "Confidence Wrapper" that upgrades standard output into a premium, authoritative tone.

Query Normalization: Implemented a layer that standardizes mixed Arabic/English inputs (e.g., automatically mapping "patrol" to "نيسان باترول").

3. AI Service Prompt Refactoring (



src/lib/ai/ai-service.ts)

Overhauled the primary system prompts to align with the new architecture.

Role Transformation: The AI is now instructed to be a "reasoning engine" rather than a copywriter.

Data-First Focus: Prompts now prioritize structured data extraction over conversational fluff.

4. Deterministic Chat Orchestration (



src/app/api/chat/route.ts)

Refactored the primary chat API route to handle the orchestration between AI reasoning and the Arabic Engine.

Intercept Flow: Arabic queries now bypass traditional AI streaming in favor of the engine's clean, premium responses.

Intent-Based Routing: The system intelligently switches between search, browse, and sell flows based on extraction.

5. Seeding & Deployment Fixes

Pre-orchestrated the production environment for a full data seed.

Full Seeding: Modified **scripts/seed_q84.ts** to allow for the full import of **579 listings** from the **q84sale** dataset, including vector embeddings for semantic search.

Docker Optimization: Updated **scripts/docker-entrypoint.sh** to prevent permission errors during the standalone build process on the production server.



6. Production Readiness

Verified the connection and verified the state of the `DigitalOcean production server (174.138.84.180)`.

Checked and confirmed that all Docker containers (`app`, `worker`, `postgres`, `qdrant`, `redis`) are operational.

Results for Today

The platform now sounds like a native, premium application for Gulf users. Responses are no longer "translated from English" but are built from the ground up to be concise, confident, and sophisticated.

Next Steps: We are ready to move these localized changes to the production server and trigger the final database seed.